

One Timeline, Many Renderings: A Wolfram Language Paclet for Heterogeneous Musical Output

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Abstract. Algorithmic composition rarely ends in a single medium: the same musical plan may need to become a Csound score for synthesis, engraved notation for performers, a real-time control stream, and a click track for rehearsal. Authored or repaired separately, these timelines drift apart. We present Temporal System, a Wolfram Language paclet that treats heterogeneous output as a projection problem: one immutable store of typed musical entities, placed on a rational beat timeline, is compiled by per-backend renderers into Csound (electroacoustic and synthesis), MusicXML (symbolic score, in beta against MusicXML 4.0), and OSC (a pre-rendered control stream for real-time dispatch) artifacts that stay synchronous because they share one store. We describe the design decisions, the anatomy (temporal, semantic, and rendering layers; fourteen modules), and the renderers. Particular attention goes to Csound, which fixes the rendering idiom: compiled tempo maps yield onset and duration seconds only at render time, session tuning yields p4 frequencies, curve entities compile to k-rate control signals bound to score p-fields, and orchestras remain ordinary `.orc` files addressed by stable named instruments—the score–orchestra separation Csound inherits from the MUSIC N family. A click-track backend derives rehearsal audio from the same meter and tempo and reuses the Csound serializer—how cheaply a further projection follows once the temporal core is shared.

Keywords: Csound, algorithmic composition, music representation, Wolfram Language, click track

1 Introduction

Long before rendering became plural, computing composers had already separated structure from sound. At Bell Labs, Tenney generated note lists for the MUSIC N programs [2,3]; in Utrecht, Koenig’s *Project 1* and *Project 2* computed a piece as tables realized at a later stage—players, the studio, and in his last works Csound itself [1]. In both, the program owns *time and structure* and a separate system owns *sound*: the orchestra–score split Csound institutionalizes [4].

Contemporary practice multiplies the realization stages: a mixed work may need a `.csd` for synthesis, MusicXML for engraving, a timed control stream for live electronics, and a click track for performers. These encode time differently—seconds, divisions per quarter, milliseconds, samples—so maintaining them by hand means maintaining four diverging timelines. Computer-assisted composition environments have long addressed parts of this [5,6]; Temporal System [15] makes one authored timeline the only source of truth and demotes every output file to a derived artifact.

One model feeding many outputs is well established. *music21* exports a single object model to MusicXML, MIDI, Lilypond and text [7]; *abjad* drives Lilypond from a persistent Python score model [8]; MEI encodes a canonical score that Verovio projects to many targets [9,10]; *bach* and *cage* carry a score/timeline abstraction into Max for real-time realization [11]. These are organized around notation or a single real-time host. Exact rational time is not new on its own either: *TidalCycles* represents patterns as functions over cycles of exact rational time [12], but in the service of live-coded pattern scheduling dispatched to a synthesis host, not of offline serialization to heterogeneous files. Temporal System’s contribution is the *combination*: one beat-space substrate of exact Rationals (musical, not clock, time) that simultaneously feeds several restrictive, heterogeneous serializers, with conversion to seconds, hertz and samples deferred to each renderer. The host language makes this practical: store, tempo-map inversion, session tuning and per-backend contracts are all Wolfram Language, in which rationals stay exact, transformations compose, and a vast built-in library (including the paclet’s own timeline graphics) sits where the timeline is authored—so authoring the piece and compiling its every projection share one language, not a chain of programs.

Temporal System’s design decisions are few and strict: (i) events live in an immutable store of typed entities whose onsets are exact Rationals in beat space, never seconds; (ii) meter is authored state, never inferred from durations; (iii) reference tuning is a session parameter, never a constant in conversion code; (iv) the packet generates *scores only*—synthesis instruments remain external `.orc` files, as in Koenig’s and Tenney’s workflows; (v) the shared store is permissive, while each renderer enforces a restrictive contract for its own payload.

One trade-off is worth stating. Authoring requires Wolfram Language, a commercial environment (a free Wolfram Engine covers development and non-commercial use), whereas Csound and much of its ecosystem are open-source. The dependency is confined to authoring: the emitted artifacts—the `.csd` with its `#included .orc` files, MusicXML, and the OSC JSON—are plain text in open formats needing no Wolfram runtime to compile, engrave, or dispatch.

2 Anatomy of the Packet

The packet comprises fourteen modules exposing 80 public functions in three layers: **core** (entities, store, tempo, session), **rendering** (dispatcher and four backends), and **util** (pitch, rhythm, dynamics, validation, timeline graphics). Layers I (temporal) and II (semantic) are backend-agnostic; Layer III holds one restrictive contract per backend, and adding a backend adds a contract there without ever touching Layers I–II. All functions are pure; store operations return new stores.

Layer I. Tempo is authored as a *tempoSpec*, an ordered list of {beat, BPM} points, and compiled to a piecewise-linear *tempoMap* in {beat, seconds} form with exact analytical inversion. Onsets and durations are Rationals in musical duration space; conversion to seconds or samples happens only inside renderers.

Layer II. The store groups entities by type (`stemData[type][id]`, $O(1)$ typed access; rendering is linear in the event count, so the model scales to large scores) across seven implemented entity types (note, rest, trigger, curve, state, marker, annotation), with four more (process, interval, trajectory, gesture) declared for future use. A note carries pitch (spelling plus cents, MIDI float, or frequency), duration, dynamic, and a **rendering** association keyed by backend id—the one place backend-specific payloads travel with the entity. Meter is a state entity: measure structure is authored, not guessed. A session configuration carries everything a renderer may not hardcode, starting with `tuningA4` (default 440 Hz, freely reassignable).

Layer III. Constructors validate only the shared shape and require each `rendering[backend]` payload to be an association whose keys belong to the backend contract; malformed Csound payloads are therefore Csound contract errors, local to one renderer, and each backend declares a fallback policy (warn, approximate) for material it cannot express.

The packet also inspects itself: `drawTimeline` renders any `stemData` as lane-based graphics with the metric grid derived from the authored meter entities (Fig. 1), from a single function call with no external tool. Every fixture in the Csound suite renders to a `.csd` that Csound compiles without error, and its rendered audio was auditioned.

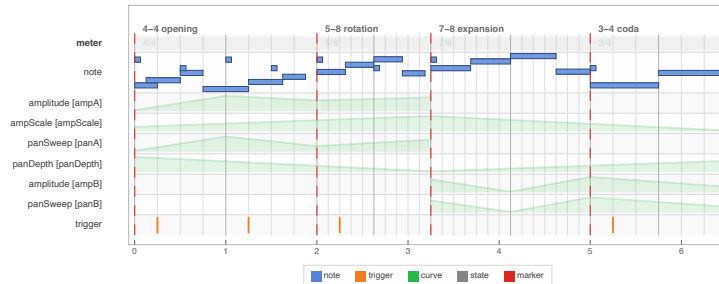


Fig. 1. A timeline exported directly by the packet (`drawTimeline`): the Csound suite’s most complex fixture—sixteen `sine_osc` notes (several with cent offsets), eight `click` hits, four triggers, six overlapping amplitude/pan curves, section markers, and a $4/4 \rightarrow 5/8 \rightarrow 7/8 \rightarrow 3/4$ meter grid. A stress fixture exercising many entity types at once, not a representative composition.

3 The Implemented Renderers

A backend is data: an association with an id, a renderer, a serializer, the primitive types it emits, and a fallback policy. The generic dispatcher `renderTo[stemData, tempo, backend, config, registry]` is the single entry point. Four backends exist. *Csound* compiles the whole `stemData` to score and orchestra primitives (Sect. 4); *Click* derives a standalone rehearsal track from meter and tempo alone (Sect. 5). *MusicXML* builds measures from meter state, splits notes at barlines into tied fragments, and validates measure capacity, part coverage, and tie and chord integrity; it is an explicitly lossy projection with a declared coverage profile [13], currently in beta, emitting MusicXML 4.0 (partwise) and checked by round-tripping through Dorico for tie, meter and part-separation fidelity. *OSC* emits a schedule-ready JSON event list in milliseconds—curves sampled at a configurable control rate—for dispatch as OSC messages [14] to whatever is listening, a graphical patcher or a running Csound instance through its OSC opcodes. Not every entity exists everywhere: curves become k-rate control signals in Csound but have no notation image; spelling matters to MusicXML, not to synthesis—differences the packet records per entity-backend pair rather than pretending projections are lossless.

4 The Csound Renderer

Csound is the oldest backend and fixes the packet’s rendering idiom, carrying the electroacoustic and synthesis output (Fig. 2). Its p-field convention is stable: **p1** instrument, **p2** onset seconds, **p3** duration seconds, **p4** frequency in Hz, **p5** amplitude in $[0, 1]$, **p6+** from the entity’s `rendering["csound"]` payload. The renderer (`renderCSnd`) takes the whole `stemData`: notes and triggers become `i` statements, markers onset-ordered score comments, and each curve a k-rate control signal in a global variable—an `f`-table reader or a generated `transeg`, per session choice. In the `f`-table form the curve is a GEN-2 table (512 points by default, a session parameter) read at k-rate by `tablei` with linear interpolation.

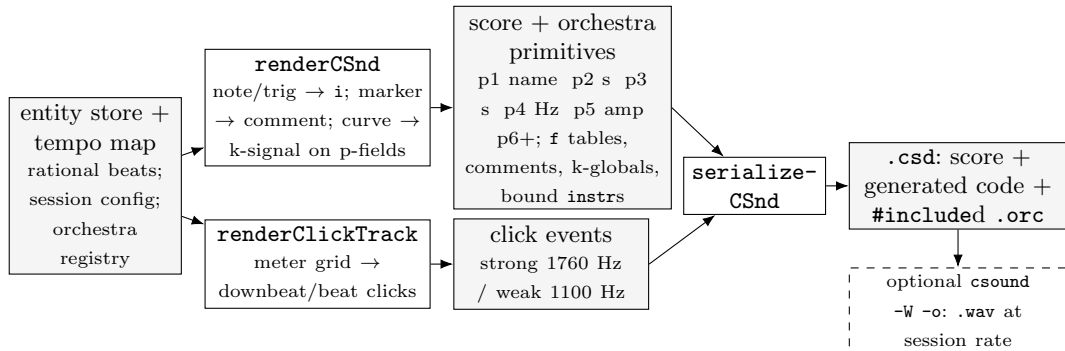


Fig. 2. Both Csound-family renderers share one serializer; source instruments stay in external `.orc` files, and curve bindings add generated copies without touching them.

Two decisions matter most. First, *late conversion*: because onsets stay Rational until this point, `p2/p3` are computed from the compiled tempo map, `p4` from spelling and cents under the session’s `tuningA4`, and `p5` from the dynamic marking, all at render time. Retuning a session to 432 Hz or editing its tempo specification regenerates every second and every frequency without touching one entity. Second, *named instruments*: an orchestra registry is loaded from a directory of `.orc` files, the filename being the stable id used both in `instr` definitions and in entity payloads. No numeric renumbering ever occurs, so the note

```
i "sine_osc" 0. 4. 440. 0.65 0.5
```

(A4, *mf*, two whole notes at 120 BPM, actual packet output) remains readable and stable as the orchestra grows.

Curves reach the sounding notes through declared bindings: by default an `amplitude` curve multiplies `p5`. For each affected instrument the renderer copies the whole `instr` body from its source `.orc`

into a *bound copy*, rewriting only the use-sites of the bound p-fields—the control-signal globals multiply or replace them, everything else (`pcount()` guards, envelopes, `limit()` clamps, the oscillator) reproduced verbatim. Affected notes are re-pointed to this copy, and several curves may compose on one p-field (in Fig. 1, two amplitude curves and an `ampScale` curve into `p5`; `panSweep` and `panDepth` into `p6`). Regenerated every render and never edited at the source, the copy cannot drift. The serializer writes the globals and generated instruments beside the registry’s `#include` lines, sorts score statements by onset, and can compile the `.csd` to audio.

The same amplitude curve lives differently outside Csound. Projected to OSC it is not bound into an instrument but sampled at the session control rate into timed messages. A linear ramp over two seconds at 10 Hz becomes a JSON event list (fields `version`, `sampleRate`, `duration_ms`, and per-sample `events`), each event an OSC message—`time_ms` 1900, `address` `/param/amplitude`, `args` `[0.95]` (actual packet output, excerpted¹). The beat-space curve and tempo map are the same as in the Csound projection; one binds the curve to a score p-field, the other streams it as OSC messages in milliseconds—a genuinely non-Csound artifact, the heterogeneity the architecture exists to make cheap.

5 A Click Track from the Same Timeline

Click tracks for mixed music are commonly rebuilt by hand in a DAW whenever the score changes—precisely the divergence this architecture removes. The click backend treats rehearsal audio as one more projection: it reads only the authored meter entities and the tempo map, builds the measure grid, and emits one short `i` event per beat, accenting each downbeat. Every parameter is session configuration—beat unit (1/4), click duration (1/16), strong/weak frequencies (1760/1100 Hz), amplitudes (0.9/0.45), and pan—and the click instrument itself is an ordinary three-line `.orc` file. Because the renderer walks the same meter grid and tempo map as every other backend, meter and tempo changes are followed with no additional authoring; for two 4/4 bars at 120 BPM it emits strong `i` "click" downbeats (0 and 2 s) and weak intervening beats (full `.csd` and `.wav` in the supplementary folder). In roughly 140 lines that reuse `serializeCSnd` unchanged (Fig. 2) and touch no line of Layers I–II, the backend inherits `#include` handling and optional `.wav` compilation—showing how cheaply a further projection follows once the temporal core is shared. The genuinely different time encodings, notation’s divisions and OSC’s milliseconds, belong to the MusicXML and OSC contracts.

6 Keeping the Projections Synchronous

Because every backend reads the same rational onsets and the same compiled tempo map, the projections cannot drift apart. Take one authored timeline—two 4/4 bars then 3/4, 120 then 90 BPM—projected to all four backends. The downbeat of bar 3 is Csound `p2= 4.0s`, OSC `time_ms= 4000`, and a strong click at 4.0s, while MusicXML places it at bar 3, beat 1. Editing the second tempo segment to 60 BPM moves every later onset by the same amount in all three time-domain backends (the note at beat 9/4: 4.667 → 5.0s) while the MusicXML export is byte-identical: notation lives in beat space, and seconds are bound only at render time. This four-backend synchrony demonstration, before and after the edit, is in the supplementary material.

7 Conclusions

Temporal System contributes an architecture, not a synthesis method: an immutable rational-beat store from which Csound scores, click tracks, notation, and real-time event streams are compiled as unequal but coordinated projections, Csound being the reference renderer that makes the contract concrete. There is deliberately no MIDI backend—MusicXML covers notation, Csound fixed-media synthesis, and OSC real-time control beyond MIDI’s resolution. The packet is not yet released; its example artifacts are on Zenodo. Current work stabilizes the MusicXML backend beyond beta and adds a hierarchical generation layer above the store. Sixty years after *Project 1* and the Bell Labs note-lists, separating computed time from rendered sound still pays—it lets one timeline become many renderings.

¹ Full example artifacts (aggregated `.csd/.wav`, click track, OSC JSON, notebook, `.orc` instruments) and the four-backend synchrony demonstration are in the supplementary folder: <https://doi.org/10.5281/zenodo.XXXXXX>.

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